Capacity Eviction in Reconstruction-Trained Tokenizers

Draft of August 10, 2026 — section text frozen pending review

Note to the reviewer. This is §7 of a paper in preparation, exported verbatim from the manuscript. Numbering matches: §7.1–§7.9. The pre-registered experiment has *not* been executed, so §6 contains no results and nothing in §7 is written with knowledge of an outcome — every limitation here was known and written before any number existed. References to §3, §4, §6, §8 and Appendix C point outside this export and are bound to their manuscript numbers; Figure 7 belongs to §5 and is reproduced at the end for reference.

Ordering principle. The subsections are ordered by what we believe a referee would attack first, not chronologically and not by severity as we perceive it. The first three bound what the paper may claim *even if every pre-registered clause passes*. If that ordering is wrong — if something in §7.5–§7.9 belongs above something in §7.1–§7.4 — that is itself a useful finding.

What we would most like challenged. (i) Whether §7.1 concedes enough: the headline mechanism is measured on a synthetic fixture, and we claim the entropy match licenses the construction without claiming it reproduces real covariance geometry. (ii) Whether the required disclosure in §7.2 is sufficient, or whether a comparison with no external anchor should not be reported at all. (iii) Whether §7.3's unquantified capacity handicap is a disclosure or a hole — we state that $\Delta\mathrm{IR}(4-5)$ mixes information with capacity and do not decompose them. (iv) Whether §7.4's power discussion is honest about a minimum detectable effect two orders of magnitude above the prior. (v) Whether §7.6's argument — that signed returns are the only channel a label leak travels on — actually holds, since it is the weakest link in the validation geometry and we know it. (vi) Anything a referee would raise that is not here at all.

---

7 Limitations

The limitations below are stated at the same confidence as the contributions, and in the order a sceptical reader should apply them. Three of them — the synthetic basis of the mechanism, the absent external baseline, and the placebo's capacity confound — bound what the paper is entitled to claim even if every pre-registered clause passes.

7.1 The mechanism is measured on a synthetic fixture, not on markets

Everything in Section 3 is measured on a constructed fixture. That is a deliberate trade and it cuts both ways. The fixture's non-signal structure is entropy-matched to the real lake — a per-dimension deficit of 0.4631 nats, reproduced at 0.462 against a declared tolerance of 0.05 — and its signal dimension is planted with an exactly computable information content, which is what allows us to say that 94.4% of a known quantity was recovered rather than that a loss curve moved. It is also precisely what stops the result being a measurement of markets. We do not know the variance and covariance geometry of real trade-flow imbalance against real price well enough to assert that the fixture reproduces it; we know only that one entropy statistic matches.

The consequence is a specific one, and we hold to it under every experimental outcome: capacity eviction is a mechanism we demonstrate, not a market phenomenon we measure, and it is not this paper's headline claim. A reader who wants the mechanism established on real microstructure will not find it here. What the pre-registered ablation of Section 4 tests is the downstream question — whether real trade-flow imbalance survives the re-specified pipeline — and a positive result there would be consistent with the mechanism without confirming it, because many other differences between cells 2 and 4 could produce the same sign.

Two narrower limits inside the same fixture. We demonstrate the effect for one reconstruction objective, one quantizer family and one class of encoder. We show that *this* objective allocates by variance and covariance; we do not show that no reconstruction objective can do otherwise, and

Section 3.6 is a counterexample to the strong form — it is a reconstruction objective that carries the evicted channel, because it was given an explicit term to do so. Separately, the 94.4% recovery figure is calibrated against a single matched reference run and inherits whatever run-to-run variation that reference carries; we report it as one measurement, not as an expectation over runs.

7.2 The BSQ baseline is ours, and nothing external anchors it

Cell 1 — binary spherical quantization on OHLCV-only inputs — is our own implementation at matched bits per token. It is not an independently validated reference. The evaluation harness was originally specified to load published reference weights and reproduce their reported rank information coefficient as an external anchor for this cell. That gate was found unexecutable and dropped as binding before any cell was scored, for three separate reasons, each sufficient on its own: the reference paper publishes two rank information coefficients for the same model that differ by 2.4× (0.0431 against 0.0665) without the pre-registration saying which applies; both are measured on Shanghai Stock Exchange 15-minute equity bars, a market and a frequency this study firewalls out of v1, so the gate's two requirements — reproduce the published number, and do it on our own pinned slice — cannot both hold; and executing the weights would require importing the reference model code, which our independence invariant forbids. Resolvability was never the obstacle: the band is 4.51 to 11.05 standard errors wide at the measured effective sample size of 1,401,637.

Two compensating controls were armed in its place, and it is important to be exact about what they do. A saturation check asks whether cell 1 was trained enough to be a fair baseline. A codebook utilization gate, pinned at 0.95, asks whether our BSQ codebook collapsed. Neither answers the question the external gate was for. We therefore carry the following sentence as a required field of every verdict manifest, refused if absent, and reproduce it verbatim here:

> We therefore cannot exclude that our BSQ baseline is weaker than a reference BSQ implementation, which would inflate the FSQ-vs-BSQ comparison reported in Section 6.

This bounds the quantizer leg specifically. It does not bound the input-arm leg: the comparison between cells 2 and 4, and between cell 4 and the placebo, is internal to our own implementation on both sides, and a weak baseline cannot inflate a difference in which the same baseline appears twice.

7.3 The placebo removes information and capacity together

The shuffled-microstructure cell is the design's central control, and its reach is narrower than its name suggests. Measured on the fixture, the shuffle destroys exactly what it was built to destroy: microstructure's contemporaneous link to price falls from 0.2037 to 0.0005, and its own lag-1 autocorrelation from 0.4165 to 0.0006, while the within-block and within-OHLCV correlations are left as they were.

But microstructure dimensions in cell 4 are partly amortized against structure the code already carries, and shuffling removes that sharing along with the information. $\Delta\mathrm{IR}(4-5)$ is therefore (information) + (capacity handicap), and **we have not quantified the handicap in information-ratio units**. A sixth cell holding the microstructure block constant rather than shuffled would bracket it, and no such cell appears anywhere in the pre-registration — we have verified this by exhaustive search of that document — so it is future work, not a committed contingency we chose not to exercise. The practical consequence: a positive $\Delta\mathrm{IR}(4-5)$ is evidence that microstructure carries information beyond input capacity, and is not a calibrated estimate of how much.

7.4 The effect we can detect is larger than the effect we expect

At the headline horizon the paired minimum detectable effect is 3.518 annualized information ratio, against an economic materiality floor of 0.5. The only prior we hold for the quantity under test is our own univariate liveness screen of trade-flow imbalance on a single symbol: $|\mathrm{RankIC}| = 0.0270$ at $h = 5$ and 0.0197 at the headline $h = 15$, both with confidence intervals excluding zero. That screen's own source describes it as a liveness screen and not the headline test, and we use it here only to set expectations about magnitude. An effect of that size does not become an information ratio of 3.5. We therefore state plainly, before seeing any result, that a null or inconclusive outcome is the most likely honest one, and that this was pre-committed as publishable: the outcome taxonomy has three members, and the decision rule names what we report under each.

A second, less visible power limitation. The minimum detectable effect above is built from *scoring* noise — the dispersion of the portfolio return series — and contains no term for training variance. That omission is not conservative, and we found its magnitude by accident: two runs at the same seed, the same recipe, the same horizon and the same estimator produced fraction-negative 0.5662 and 0.99883 for the same cell. Same-seed divergence is basin-hopping, and it is a lower bound on across-seed variance, which is the quantity the design actually averages over. Five seeds are pinned up front, which helps; forcing deterministic kernels does not help, because it removes the same-seed component and leaves the across-seed component untouched.

What we do about it is a refusal rather than a correction. A power guard, computed after the clauses and unable to alter them, compares each cell's across-seed range against the effect being claimed; if the range reaches the claimed effect, the verdict is reported as HALT_ADJUDICATE rather than as an outcome. The guard is HALT-only by construction — it can manufacture neither a positive nor a negative claim — which is what makes it legitimate to have added it before the run. It does not increase power. It prevents us from reporting a number smaller than its own inputs' wobble.

7.5 One exchange, one venue, one year of scored data

The lake is 304,625,181 one-minute bars across 200 instruments, and the breadth of that number should not be read as breadth of evidence. Every instrument is a USDT-margined perpetual future on a single exchange. Perpetuals are not spot, and one exchange is not the market: fee schedules, tick sizes, liquidation mechanics and the composition of flow all differ across venues, and trade-flow imbalance is a venue-local quantity in a way that returns are not. Nothing here establishes that the result transfers to spot, to another exchange, to dated futures, or to any other asset class.

Within that universe the scored slice is narrower still. The headline metric is computed on 40 instruments, not 200, and on forward blocks 1 through 5 — roughly 2024-01-01 to 2025-01-01, one year — with block 0 spent selecting the execution filter and excluded. The remaining 160 instruments are lake breadth, not evaluation breadth. One year of scored data at a 15-bar horizon is 35,063 periods per instrument, and the effective breadth across a correlated 40-instrument cross-section is 1.5, not 40 — which is where most of the minimum detectable effect in Section 7.4 comes from. That single year is also one market regime. We do not test regime transfer and cannot claim it.

Survivorship is owned rather than filtered: instruments that stop trading are retained, and 17 of them stop inside the window, giving the ragged right edge of Figure 7. This removes one bias; it does not remove listing bias, since an instrument must have been listed on this venue to enter the universe at all.

7.6 The embargo premise is established for one channel only

The 120-bar leading embargo is a 60-bar label look-forward plus a 60-bar serial-correlation margin, and the premise behind the second margin was measured on all 200 instruments rather than assumed, with a control arm that recovers a known AR(1) answer first so that a null reading is a measurement and not a silent failure. For signed returns it holds comfortably: autocorrelation at lag 60 is 0.0067 on average and 0.0191 at the worst instrument.

For absolute returns it does not hold: 0.2323 on average and 0.3604 at the worst instrument, at the same lag. We report this as a red finding rather than a footnote. Our position is that signed returns are the channel a label leak would actually travel on — the labels are signed forward returns, and a leak means a training window seeing the sign of a future return it should not — and that persistent absolute-return autocorrelation is volatility clustering, a well-established property of the series that is present in the training region and the evaluation region alike and is not a path from label to feature.

That is an argument, and it is the weakest link in the validation geometry. What we have established is that the signed channel is clean at the chosen margin. What we have *not* established is that no leakage path exists through volatility — for instance, a model that learns to condition on realized volatility could in principle be advantaged by a training window whose volatility state persists into the evaluation region. Extending the margin until the absolute-return premise also holds would require a margin several times longer and would cost evaluation data we do not have; we chose the shorter margin and we are stating the choice.

7.7 The microstructure leg is narrower than "microstructure" suggests

Trade-flow imbalance, not order-flow imbalance. The imbalance signal is computed from freely available aggregated trades, and is therefore signed *executed-volume* imbalance. True order-flow imbalance requires order book depth — the arrival and cancellation of resting liquidity — which we do not have. The distinction is not cosmetic: order-flow imbalance carries information about intent that never executes, and a substantial part of the microstructure literature's predictive content comes from exactly that. We use the name trade-flow imbalance everywhere, in code, configuration and prose, and a negative result here is evidence about trade-flow imbalance only. Depth-based order-flow imbalance is explicit v2 work.

Three of sixteen feature dimensions are structurally absent. The per-bar feature vector is specified as 16 dimensions, of which 13 are live. Funding rate and open interest occupy dimensions 13 through 15; they are wired, carried through the pipeline, and masked — and in the v1 lake they have standard deviation exactly 0.0 and are masked on 100% of all 304,625,181 bars, because ingest covers klines and aggregated trades only. They are specified and structurally absent. The microstructure block that the ablation actually varies is six live dimensions, 7 through 12, each non-degenerate across all 200 instruments. Funding and open interest are the two channels most plausibly carrying the information a perpetual-futures microstructure study would want, and the study is run without them. This is a limitation of the v1 data, not a design choice defended on merit.

7.8 Two matching caveats

λ is fitted to a synthetic proxy. The class weight that makes the per-bar bottleneck carry the microstructure block was selected on the calibration fixture, not on real data, and its acceptance threshold was restated after calibration results were seen — we state that plainly in Section 3.6 rather than let it be discovered. It is pinned pre-data at a single value and applied identically to every arm, so it cannot favour any cell relative to any other. The residual risk is that the proxy does not transfer: a λ tuned for a synthetic covariance geometry may under-serve the real one. The guard is the legibility gate, which is why it exists — it is evaluated on real data before any second-stage training begins, requires at least 0.90 sign accuracy on every one of the six live microstructure dimensions with no mean clause and no floor, and raises a hard stop rather than degrading quietly. A gate that fires costs us the run; it does not cost us the honesty of the result.

The arms are matched to 0.327%, not exactly. The backbone realizes 21,301,248 parameters at the FSQ vocabulary and 21,231,616 at the BSQ vocabulary — 69,632 fewer, or 0.327% — because embedding and output-head rows scale with vocabulary size. The two quantizer arms also differ by 0.058 bits per bar. Both gaps favour the FSQ arm, both are an order of magnitude smaller than any effect the design can detect, and both are reported rather than absorbed. We do not claim exact parameter parity; we claim parity to a stated tolerance, hard-checked before any cell is scored.

7.9 Reproducibility is bit-exact except in one place, and that place is training

The data pipeline, the frozen normalization statistics and prediction replay are bit-exact from one configuration file, a pinned seed and content-hashed inputs. GPU training is not, unless the deterministic-attention fallback is enabled, and the two are not the same claim: deterministic attention is necessary for bit-exact training and is not sufficient, because reduction order elsewhere in the backward pass is unconstrained. We record the mode for every run. Section 8 states exactly which artifacts carry which guarantee, and Appendix C separates what was measured from what was estimated across the whole paper.

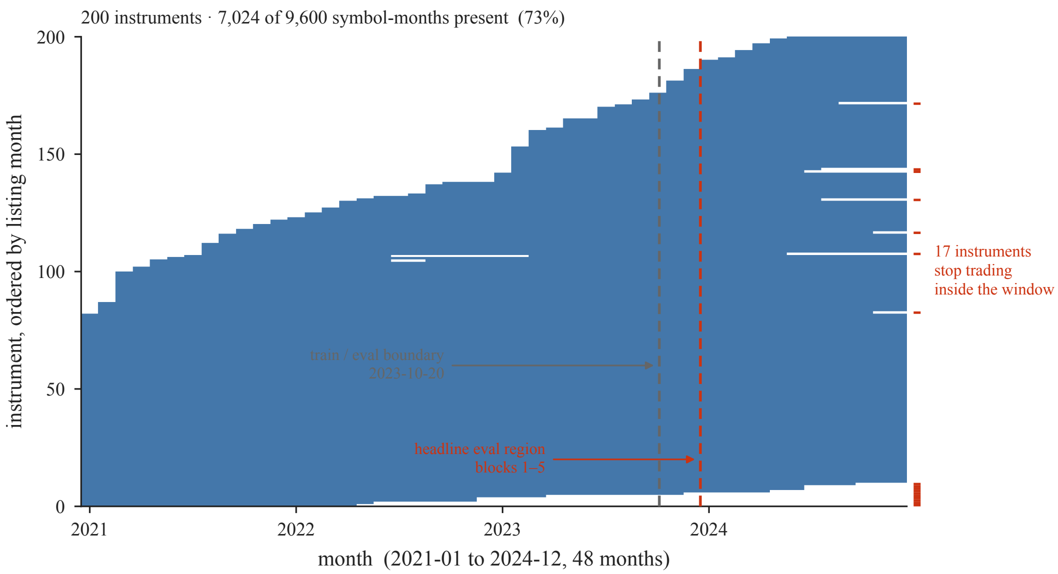

Figure 7: Reproduced from §5 for reference only — Section 7.5 points at this panel. Month-by-month presence for all 200 instruments over the four-year window, ordered by listing month: 7,024 of a possible 9,600 symbol-months are present. The right edge is ragged because instruments that stop trading are *kept*, and 17 of them stop inside this window.

References